

Annotated Bibliography

Medical AI Paper & Dataset Publications — LinkedIn Sample

Scope: Medical AI / deep learning for medical imaging & diagnostics, sourced from LinkedIn content search (same field as the reference post on skin cancer classification, *Diagnostics* MDPI). Window: **January 2025 – September 16, 2026**.

Total entries: 21

Note: This is a sample compiled from LinkedIn content search, not an exhaustive census — LinkedIn's search interface does not expose a countable total of matching posts. Results below are what a structured set of targeted searches (under an authenticated LinkedIn session) surfaced and manually screened for recency and relevance, not every qualifying post that exists.

1. Benchmarking Class Imbalance Mitigation Strategies Across Deep CNN Architectures for Skin Cancer Classification

Authors & Journal Name: Dr. Irshad Ahmad, Muhammad Khubaib, Saleh Altowaijri; Journal: *Diagnostics* (MDPI); (Peer-reviewed, Open Access)

Country and Dataset: Country: Pakistan / Saudi Arabia (Smart Technologies Lab, Islamia College Peshawar); Dataset: ISIC 2019 (skin lesion image dataset)

Methodologies: Benchmarked 16 class-imbalance mitigation techniques (incl. Balanced MixUp) across 6 pretrained deep CNN architectures: EfficientNet-B0, EfficientNet-B3, ResNet50, DenseNet121, InceptionV3, and MobileNetV2, for skin lesion classification.

Summary Findings: Balanced MixUp combined with EfficientNet-B3 achieved the best result: 92.39% accuracy and 92.33% F1-score, showing that the choice of imbalance-mitigation strategy matters as much as the choice of architecture. Reference post the user originally linked to for this exercise.

Variables: Dependent variable;; Skin lesion class (multi-class skin cancer diagnosis); Independent variables;; Dermoscopic image (ISIC 2019); CNN architecture used; Class-imbalance mitigation technique applied

2. Attention-Guided EfficientNet-B3 with Grad-CAM Visualization for 22-Class Bone Fracture and Anatomical-Region Classification on the MultiBoneX Dataset

Authors & Journal Name: Mian Hafeez, Dr. Irshad Ahmad, Saleh Altowaijri; Journal: *Diagnostics* (MDPI), Vol. 16, Issue 17, Article 2841; DOI: 10.3390/diagnostics16172841

Country and Dataset: Country: Pakistan / Saudi Arabia; Dataset: MultiBoneX Dataset (22-class bone fracture/anatomical-region imaging)

Methodologies: Attention-guided EfficientNet-B3 with CBAM (Convolutional Block Attention Module), combined with Grad-CAM for visual interpretability, applied to 22-class bone fracture and anatomical-region classification.

Summary Findings: Achieved 75.03% test accuracy on the 22-class task. Grad-CAM visualizations were used to support interpretability of model predictions for clinical trust.

Variables: Dependent variable;; Bone fracture / anatomical-region class (22 classes); Independent variables;; X-ray/CT image (MultiBoneX); CBAM attention weights; EfficientNet-B3 feature representations

3. A Preliminary Comparison of EfficientNet-B3 and ResNet-50 for Detection and Subtype Classification of Intracranial Hemorrhage on Head Computed Tomography

Authors & Journal Name: Awais Jan, Maryam Sajjad, Muhammad Hammad Hassan Mallick, Leena Faraz, Muhammad Mohsin, Bushra Ashar; Journal: *Journal of Saidu Medical College (JSMC)*

Country and Dataset: Country: Pakistan (Frontier Medical College Abbottabad; GIKI Sawabi; NUST Islamabad; UET Mardan; Swat Medical College; Dow University of Health Sciences Karachi); Dataset: Head CT scan images (not further

specified)

Methodologies: Compared EfficientNet-B3 and ResNet-50 deep learning architectures for detecting and classifying subtypes of intracranial hemorrhage (ICH) from head CT scans.

Summary Findings: Reported high patient-level sensitivity and ROC-AUC scores, supporting the feasibility of deep learning-assisted ICH subtype detection in medical imaging diagnostics.

Variables: Dependent variable:; ICH subtype presence/class; Independent variables:; Head CT scan image; Model architecture (EfficientNet-B3 vs. ResNet-50)

4. Lung Cancer Detection Using Medical Image Processing

Authors & Journal Name: Paarth Kapur, Alekhya Gorugantu, Rujuta Ashtekar (mentored by Dr. Nandana Prabhu, Dr. Renuka Ashtekar); Journal: Springer, Proceedings of ICAIN 2025 (Lecture Notes in Networks and Systems, Vol. 5), book chapter; DOI: 10.1007/978-3-032-23314-1_12

Country and Dataset: Country: Not specified; Dataset: CT scan lung cancer imaging dataset (4 classes: Adenocarcinoma, Large Cell Carcinoma, Squamous Cell Carcinoma, Normal)

Methodologies: Compared DenseNet201, ResNet50, a hybrid DenseNet201-ResNet50 architecture, and EfficientNet-B3 (via transfer learning) for 4-class lung cancer classification from CT scans.

Summary Findings: The EfficientNet-B3 transfer-learning model achieved the best result: 94.29% test accuracy across the four lung cancer classes.

Variables: Dependent variable:; Lung cancer class (4 categories); Independent variables:; CT scan image; Model architecture used

5. Multi-approach technique implementation for segmentation of left ventricular myocardial infarction and quantification

Authors & Journal Name: Dr. Kavitha K.S. (Dept. of CSE, Dayananda Sagar College of Engineering); Journal: Scientific Reports

Country and Dataset: Country: India (Bangalore); Dataset: Cardiac imaging data (not further specified)

Methodologies: Combines image preprocessing, deep learning, and optimisation-based approaches for segmentation of left ventricular myocardial infarction.

Summary Findings: Demonstrates improved identification and measurement (quantification) of myocardial infarction from cardiac medical images, illustrating computational intelligence's potential in advanced medical image analysis.

Variables: Dependent variable:; Segmented myocardial infarction region / quantification metric; Independent variables:; Cardiac image; Preprocessing method; Optimisation-based segmentation parameters

6. Artificial Intelligence in Healthcare Practice: Validation, Fairness, and Regulatory Challenges

Authors & Journal Name: Ghulam Hussain Noori, Shaista Bibi, Seung Won Lee; Journal: INQUIRY: The Journal of Health Care Organization, Provision, and Financing (Sage)

Country and Dataset: Country: Not specified; Dataset: N/A (systematic review of 20 clinical AI studies, 2020-2025)

Methodologies: Systematic review of 20 clinical AI studies (2020-2025) assessing validation practices, fairness, and regulatory readiness of clinical AI, with a focus on diagnostic imaging applications.

Summary Findings: Deep learning (especially CNNs) dominates medical imaging diagnostics, with a median AUC of 0.91. Top clinical domains are Radiology (30%), Oncology (20%), Cardiology (15%). Main adoption barriers: regulatory compliance (55%), limited algorithmic transparency (40%), data quality limitations (35%), clinical workflow integration (30%).

Variables: Dependent variable:; Clinical AI adoption / validation outcome (qualitative, review-level); Independent variables (review dimensions):; Clinical domain; Validation practice type; Fairness assessment presence; Regulatory compliance status

7. An Intelligent Gene Biomarker Identification Framework for Pan-Cancer Classification using Genetic Algorithms and Deep Learning

Authors & Journal Name: Aruja Tiwary (1st author), Lubhana Mutha, Daksh Patil, Dr. Hima Deepthi Vankayalapati, Prof. Prateeksha Shanoj; Journal: IEEE, Proceedings of the International Conference on Bioinformatics and Computational Biology (ICBCB 2026); DOI: 10.1109/ICBCB69424.2026.11621848

Country and Dataset: Country: India (IIT Kanpur); Dataset: TCGA (The Cancer Genome Atlas) - 10,459 patient samples, 33 cancer types

Methodologies: Used a Genetic Algorithm to select an informative gene subset from TCGA gene-expression data (avoiding the full transcriptome), then trained two attention-based models, TabNet and SAINT, on that subset.

Summary Findings: SAINT reached 98.76% accuracy on cancer detection and 95.60% across all 33 tumor types. Gradient attribution analysis showed the model relied on biologically meaningful genes (e.g. EGFR/KRAS for lung cancer, BRCA1/ESR1 for breast cancer, APC/TP53 for colorectal cancer), confirming genuine oncology signal rather than noise. Shared separately by two co-authors on LinkedIn; counted once.

Variables: Dependent variable;; Cancer type (33-class) / cancer vs. non-cancer; Independent variables;; Gene expression profile (TCGA); Genetic-Algorithm-selected gene subset; Model type (TabNet vs. SAINT)

8. Cross-Domain Transfer Learning and Self-Supervised Representation Learning for Automated Histopathological Image Classification in Colorectal Cancer Grading System

Authors & Journal Name: Jeya Sherlyne P; Journal: IEEE

Country and Dataset: Country: Not specified; Dataset: Histopathological image dataset (not further specified)

Methodologies: Combines cross-domain transfer learning, self-supervised representation learning, patch-level analysis, domain adaptation, synthetic data generation, a Swin Transformer backbone, multimodal learning, and Grad-CAM explainability for colorectal cancer grading.

Summary Findings: Framework is intended to improve accuracy, generalization, and interpretability of automated histopathological cancer grading. Author's second IEEE publication; her first (ConvNeXtV2 for skin lesion classification) is referenced but not separately dated/counted here.

Variables: Dependent variable;; Colorectal cancer grade (from histopathological image); Independent variables;; Histopathological image patch; Domain-adaptation/self-supervised features; Swin Transformer representations

9. A Comparative Experimental Study of DenseNet121, EfficientNetB7, MobileNetV2, and ConvNeXTV2 for Breast Cancer Classification

Authors & Journal Name: HirenKumar Kukadiya & Divyakant Meva (highlighted by International Journal of Advanced Trends in Computer Applications, IJATCA); Journal: IJATCA

Country and Dataset: Country: Not specified; Dataset: Mammographic imaging dataset (not further specified)

Methodologies: Comparative experimental study of four deep learning architectures (DenseNet121, EfficientNetB7, MobileNetV2, ConvNeXTV2) for breast cancer classification from mammographic images.

Summary Findings: Presents a head-to-head comparison of the four architectures' performance for breast cancer classification (specific accuracy figures not disclosed in the LinkedIn summary).

Variables: Dependent variable;; Breast cancer class (malignant/benign or subtype); Independent variables;; Mammographic image; Model architecture used

10. AI-Driven Multimodal Deep Learning Framework for Automated Lung and Skin Cancer Detection and Classification from Medical Images

Authors & Journal Name: Palagummi Syam Kumar, Namadi Jeevana Dheepa, Thummalapalli N S S Siddhardha, Bokka Ganapathi, Bogabattina Venkata Manikanta; Journal: International Journal for Research Trends and Innovation (IJRTI), Vol. 11, Issue 8; Paper ID: IJRTI2608060

Country and Dataset: Country: Not specified; Dataset: Medical imaging data (lung and skin cancer; not further specified)

Methodologies: Multimodal deep learning framework combining medical image analysis, feature learning, and classification for automated detection of both lung and skin cancer from medical images.

Summary Findings: Positioned as supporting faster, more consistent, technology-driven healthcare decision-making for lung and skin cancer diagnosis; specific performance metrics not disclosed in the LinkedIn summary.

Variables: Dependent variable:: Cancer type/class (lung or skin); Independent variables:: Medical image (multimodal input); Deep learning architecture used

11. Skin Cancer Classification Using Deep Learning Techniques

Authors & Journal Name: Vandana H, Arpitha V G, Anvitha GH, Neha Khanum (guided by Dr Uma Mahesh R N); Journal: IEEE Xplore, Annual International Conference on Data Science, Machine Learning and Blockchain Technology (AICDMB-2025)

Country and Dataset: Country: India (ATME College of Engineering); Dataset: Not specified

Methodologies: Mini-project applying deep learning techniques for skin cancer classification, presented at AICDMB-2025.

Summary Findings: Demonstrates feasibility of a student-led deep learning pipeline for skin cancer classification (specific accuracy figures not disclosed in the LinkedIn summary).

Variables: Dependent variable:: Skin cancer class; Independent variables:: Skin lesion image; Deep learning model used

12. Skin Cancer Classification using Machine Learning

Authors & Journal Name: Kavya S., Himanshu Chauhan, Krishna Kumar (mentored by Dr Dhyanendra Jain); Journal: IEEE Xplore, International Conference on Innovative Practices in Technology and Management (2026)

Country and Dataset: Country: Not specified; Dataset: Dermoscopic image dataset (not further specified)

Methodologies: Web-based deep learning system for skin cancer classification from dermoscopic images, using a CNN, deployed with TensorFlow.js and a React web application for real-time, in-browser predictions.

Summary Findings: Achieved ~94.28% accuracy classifying 7 types of skin lesions; designed to make early detection faster, more accessible, and privacy-friendly by running predictions directly in the browser.

Variables: Dependent variable:: Skin lesion type (7 classes); Independent variables:: Dermoscopic image; CNN model (TensorFlow.js)

13. Multi-Class Classification of Cervical Cancer Cells Using Deep Learning Techniques

Authors & Journal Name: E. Anisha Patra; Journal: TANZ Research Journal

Country and Dataset: Country: India (NIST University, Brahmapur, Odisha); Dataset: Cervical cancer cell imaging dataset (not further specified)

Methodologies: Applies deep learning techniques to multi-class classification of cervical cancer cells to improve accuracy and efficiency of early detection.

Summary Findings: Contributes to improved cervical cancer cell classification accuracy for early detection and better healthcare outcomes (specific performance metrics not disclosed in the LinkedIn summary).

Variables: Dependent variable:: Cervical cancer cell class; Independent variables:: Cell imaging data; Deep learning model used

14. An Optimized Framework for Accurate Skin Cancer Classification

Authors & Journal Name: Parthasarathy M and co-authors; Journal: IEEE Xplore, 2025 IConSCEPT conference

Country and Dataset: Country: Not specified; Dataset: Not specified

Methodologies: Hybrid deep learning approach integrating a DenseNet201 backbone with a custom CNN architecture to improve detection of malignant skin conditions.

Summary Findings: Framework designed to overcome challenges of traditional models in enhancing detection of malignant skin conditions (specific accuracy figures not disclosed in the LinkedIn summary).

Variables: Dependent variable:: Skin condition class (malignant vs. benign); Independent variables:: Skin image; DenseNet201 + custom CNN features

15. HistAdvNet: A Robust Deep Learning Framework for Histopathology Classification Under Adversarial Conditions

Authors & Journal Name: Sazzad Hossain and co-authors; Journal: Biomedical Signal Processing and Control (Elsevier, Q1 journal)

Country and Dataset: Country: Not specified; Dataset: Lung and colon cancer histopathology image datasets

Methodologies: Introduces HistAdvNet, a novel lightweight deep learning framework for lung and colon cancer histopathology image classification, designed and tested for robustness against adversarial attacks.

Summary Findings: Demonstrates that HistAdvNet maintains strong classification accuracy while showing improved robustness against adversarial perturbations, compared to standard architectures.

Variables: Dependent variable:: Lung/colon cancer histopathology class; Independent variables:: Histopathology image (incl. adversarially perturbed versions); HistAdvNet architecture

16. Convolutional Neural Network for Lung Cancer Prediction and Classification

Authors & Journal Name: JAMI Jaswanth, Palameti Reddy Lakshmi Manoj (guided by Mrs. Sakthi U); Journal: IEEE Xplore Digital Library

Country and Dataset: Country: India (SRM University, KTR campus); Dataset: CT scan imaging dataset (not further specified)

Methodologies: Uses a Convolutional Neural Network (CNN) for prediction and classification of lung cancer from CT scan images.

Summary Findings: Aims to support faster and more accurate medical diagnosis of lung cancer using CNN-based image classification (specific accuracy figures not disclosed in the LinkedIn summary).

Variables: Dependent variable:: Lung cancer presence/class; Independent variables:: CT scan image; CNN model

17. Colon Cancer Classification from Histopathological Images Using Convolutional Neural Networks

Authors & Journal Name: Yuvashree M; Journal: International Research Journal on Advanced Engineering and Management (IRJAEM), Vol. 04, Issue 03

Country and Dataset: Country: Not specified; Dataset: Histopathological image dataset (not further specified)

Methodologies: Applies CNN-based deep learning techniques to classify colon cancer from histopathological medical images.

Summary Findings: Reports improved accuracy in colon cancer detection from histopathological images using CNN methods (specific figures not disclosed in the LinkedIn summary).

Variables: Dependent variable:: Colon cancer class; Independent variables:: Histopathological image; CNN model

18. NeuroTrustNet: A Cost-Effective Multimodal Ensemble Framework for Brain Tumor Classification Under Cross-Dataset Variability

Authors & Journal Name: Ferdaus Ibne Aziz (1st author), Tumennast E. (supervisor/co-author) and team; Journal: Frontiers in Artificial Intelligence (Impact Factor 6.7)

Country and Dataset: Country: South Korea (University of Ulsan); Dataset: Multiple external MRI datasets (cross-dataset evaluation)

Methodologies: Proposes NeuroTrustNet, a multimodal AI framework combining CNNs, Vision Transformers (ViTs), and radiomic features through Adaptive Attention Stacking (AAS), to balance performance, computational efficiency, and interpretability.

Summary Findings: Achieved 94% accuracy on a completely unseen external MRI dataset, demonstrating strong cross-dataset generalization with improved computational efficiency compared to single-model baselines.

Variables: Dependent variable:: Brain tumor class; Independent variables:: MRI image (cross-dataset); Radiomic features; CNN/ViT ensemble outputs (AAS)

19. An explainable deep learning approach for automated detection and grading of diabetic retinopathy from fundus images

Authors & Journal Name: Marisa Oishy and co-authors (incl. Muradul Islam; supervised by Ferdaus Anam Jibon, Gahangir Hossain); Journal: Microvascular Research (Elsevier)

Country and Dataset: Country: Not specified; Dataset: Fundus image dataset (not further specified)

Methodologies: Hybrid and explainable deep learning approach for automated detection and grading of diabetic retinopathy severity from retinal fundus images; originated as an undergraduate thesis.

Summary Findings: Contributes an explainable (interpretable) deep learning pipeline for diabetic retinopathy grading, intended to support clinical trust in automated grading decisions (specific accuracy figures not disclosed in the LinkedIn summary).

Variables: Dependent variable:: Diabetic retinopathy grade/severity; Independent variables:: Fundus (retinal) image; Explainability (XAI) feature maps

20. RACLETTE: Realistic Synthetic Aortic 4D/5D Flow MRI with Computational Enhancement for Training and Testing Algorithms

Authors & Journal Name: Pietro Dirix, Stefano Buoso, Sebastian Kozerke (ETH Zurich Cardiovascular Magnetic Resonance Group); Open dataset release

Country and Dataset: Country: Switzerland (ETH Zurich); Dataset: RACLETTE itself is the new dataset - CFD-enhanced synthetic MRI: 189 virtual subjects (Aortic 4D Flow MRI) + 25 virtual subjects (Aortic 5D Flow MRI)

Methodologies: Combines Computational Fluid Dynamics (CFD) simulations with MRI signal modeling to generate realistic synthetic cardiovascular MRI datasets with known ground truth.

Summary Findings: Public release of a new open dataset (RACLETTE) enabling controlled, reproducible benchmarking of image reconstruction, segmentation, and hemodynamic quantification algorithms for cardiovascular MRI research.

Variables: Output (dataset content):: Synthetic 4D/5D Flow MRI volumes with known ground truth; Input factors (dataset generation):: CFD simulation parameters; MRI encoding variant (standard, super-resolution, icosahedral)

21. COBRA2026: A large-scale pelvic cone-beam CT projection dataset

Authors & Journal Name: Adrian Thummerer and co-authors (incl. Lukas Zimmermann, Simon Rit, Florian Kamp, Matteo Maspero, Guillaume Landry, and others; 6 European radiotherapy centers); Associated with the COBRA2026 Deep Learning Challenge (MIDL 2027)

Country and Dataset: Country: Multi-country (Europe) - 6 radiotherapy centers using Elekta and Varian systems; Dataset: COBRA2026 itself is the new dataset - 867 pelvic CBCT acquisitions, ~950GB

Methodologies: Collected raw clinical CBCT projection data, acquisition geometry/calibration/correction information, clinically reconstructed CBCT images, planning CT images, deformably aligned CT references, and simulated projections across 6 European radiotherapy centers.

Summary Findings: Provides a large-scale, openly released benchmark dataset (COBRA2026) to support development and reproducible benchmarking of conventional and deep learning-based cone-beam CT reconstruction and image correction methods in radiotherapy; forms the basis of the COBRA2026 Deep Learning Challenge at MIDL 2027.

Variables: Output (dataset content):: CBCT projection data, reconstructed images, planning CTs, simulated projections;
Input factors (dataset generation):: Imaging system (Elekta vs. Varian); Acquisition geometry/calibration

Compiled via LinkedIn content search under an authenticated LinkedIn session (Sunanda Biswas). See the companion Excel workbook (Medical_AI_Paper_Dataset_Publications_LinkedIn.xlsx) and Recent Publications.md / Recent Publications.pdf for the same data in spreadsheet and summary-entry formats.